

Portfolio Builder — System Overview

Multi-tenant portfolio platform · React (Vite) + FastAPI + SQLite · Generated from codebase at [portfolio/](#)

1. What the system does

Users register with an email, password, and unique public **slug**. Each account owns **exactly one** site. They edit content in a dashboard, publish it, and visitors open `/u/<slug>`. Contact forms store messages per site.



2. Architecture (layers)

Frontend — React 19 + Vite + React Router
localhost:5173 · Pages: Landing, Login, Register, Dashboard, SitePage
Components: Hero, About, Projects, Experience, Skills, Contact, Nav, Footer...
Dev proxy: `/api` → `http://127.0.0.1:8001` (or 8000)

Backend — FastAPI (Uvicorn)
localhost:8000/8001 · Router prefix `/api`
Modules: `main.py`, `routes.py`, `auth.py`, `db.py`, `schemas.py`, `config.py`

Database — SQLite
backend/data/portfolio.db
Tables: `users`, `sites`, `sessions`, `messages`

3. Auth & identity workflow



Dashboard does *not* take a slug from the URL. The session cookie identifies the user; the DB returns that

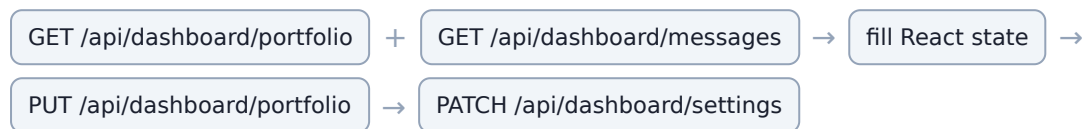
user's single site (and its slug).

4. End-to-end workflows

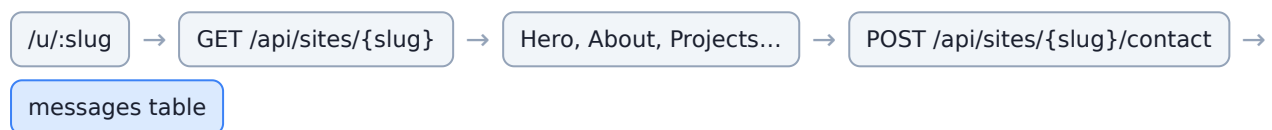
A. Registration



B. Dashboard load & save



C. Public site & contact



5. API map

Method	Path	Auth	Purpose
GET	/api/health	—	Liveness
POST	/api/auth/register	—	Create user + site + session
POST	/api/auth/login	—	Session cookie
POST	/api/auth/logout	cookie	Delete session
GET	/api/auth/me	required	Current user + site summary
GET	/api/dashboard/portfolio	required	Owner portfolio JSON
PUT	/api/dashboard/portfolio	required	Save portfolio
PATCH	/api/dashboard/settings	required	Theme / published
GET	/api/dashboard/messages	required	Inbox for owner site
GET	/api/sites/{slug}	optional	Public site (drafts: owner only)
POST	/api/sites/{slug}/contact	—	Contact form (honeypot)

6. Data model

users

- id, email (unique)
- password_hash (PBKDF2)
- created_at

sessions

sites (1 per user)

- user_id UNIQUE · slug UNIQUE
- theme (night | ice)
- published 0/1
- portfolio_json — full document

- token PK → cookie value
- user_id, expires_at

messages

- site_id, name, email, message
- created_at

7. Portfolio JSON shape (stored in sites.portfolio_json)

Key	Used by UI	Contents
profile	Nav, Hero, Contact, Footer	name, role, location, email, links...
hero	Hero	nameLines, accent, lead, pass fields
about	About	headline, emphasis, secondary
stats / marquee	Strip	highlight numbers + scrolling tags
projects	Projects	title, stack, github, live, scene
experience	Experience	role, company, points
skillGroups	Skills	grouped skills with icons
education	Skills	degree, school, period

8. Frontend routing

Path	Page	Data source
/	LandingPage	auth context
/login	LoginPage	POST login → /dashboard
/register	RegisterPage	POST register → /dashboard
/dashboard	DashboardPage	session → owner site
/u/:slug	SitePage	GET sites/{slug} → components

9. Request path (dev)



10. Key constraints

- One site per user (sites.user_id UNIQUE)
- Slugs unique globally; 3-32 chars, lowercase, numbers, hyphens
- Unpublished sites: 404 for public; owner can preview while signed in
- Contact honeypot field website — if filled, silently discarded
- Demo seed: mahi@demo.local / mahi1234 → /u/mahi

11. Module responsibilities

File	Role
frontend/src/api.js	HTTP client; credentials include
frontend/src/auth.jsx	AuthProvider; login/register/me state
frontend/src/pages/SitePage.jsx	Load public portfolio; pass props to Hero etc.
backend/app/routes.py	All HTTP endpoints
backend/app/auth.py	Hash/verify password; session cookie; current_user
backend/app/db.py	SQLite CRUD
backend/app/schemas.py	Pydantic validation
backend/app/portfolio_default.py	EMPTY + DEMO portfolio templates

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